



Design Review Report

kiln2

kiln2.zip

Report ID
2bec49e9-347b-45c1-a998-e8b8b3afe978

Report Date
July 11, 2026

Project
kiln2

Model
gpt-5.4

Revision
Initial

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July 11, 2026

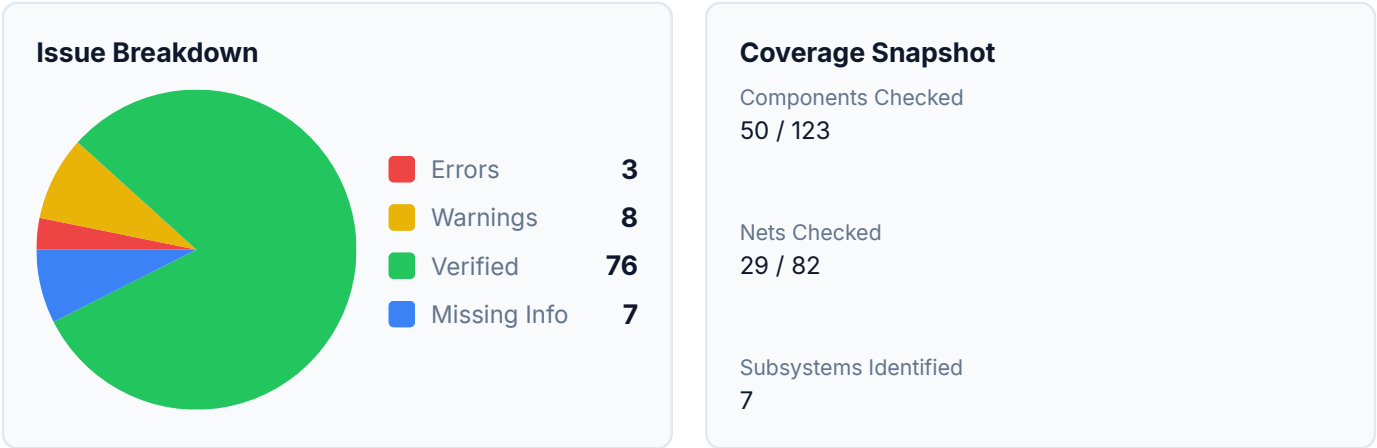
Netlist
kiln2.zip

Warning

AI can make mistakes. Findings are automated and should be validated by a qualified engineer before fabrication or release. Check important info.

Executive Summary

Merged seven review outputs covering power conversion, relay switching, safety processing, thermal sensing, analog monitoring, and cross-boundary interfaces. The main errors are a repeated relay wiring fault that can short the 12 V rails when energized, powering the Raspberry Pi Pico module through its 3V3 output pin, and a missing pull-up on the U5 optocoupler output. Additional warnings cover TVS clamp margin, 3.3 V relay-MOS-FET drive characterization, optocoupler bias/CTR margin, a MAX31856 thermocouple front-end deviation from the vendor reference, and analog input connector behavior. Several LT8631- and MAX31856-specific checks remain unresolved because the required datasheet details were not available.



Report Details

Report ID	2bec49e9-347b-45c1-a998-e8b8b3afe978	Revision	Initial
Report Date	July 11, 2026 at 08:45 PM UTC	Generated	July 11, 2026 at 08:53 PM UTC
Model	gpt-5.4	Provider	openai

Datasheets Used

- 1935161.pdf
<https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/909/19...>
- BSP296NH6327XTSA1.pdf
https://www.infineon.com/dgdl/Infineon-BSP296N-DS-v02_00-en.pdf
- EE2-12NUH.pdf
https://yageogroup.com/content/datasheet/asset/file/KEM_R7002_EC2_EE2
- MAX31856.pdf
<https://www.mouser.com/datasheet/3/1014/1/max31856evsys.pdf>
- RaspberryPi_Pico.pdf
<https://datasheets.raspberrypi.com/pico/pico-datasheet.pdf>
- SJ2-35813B-SMT-TR.pdf
<https://www.cuidevices.com/product/discontinued/sj2-3581x-smt.pdf>
- SMAJ24CA.pdf
<https://www.st.com/resource/en/datasheet/cd00001333.pdf>
- TCMT1109.pdf
<https://www.vishay.com/docs/83510/tcmt11.pdf>

Errors (3)

1. [K1/K2/K3/K4/K5] Relay contact common is tied to the coil-switch node on all five EE2 channels

Full Description

With K1/K2/K3/K4/K5 pin 9 and pin 12 connected to the same output net as J8/J9/J14/J16/J17 pad 2, each relay contact common shares the same node as the coil return and MOSFET drain. The EE2 non-latch pinout defines pins 1 and 12 as the coil, pin 9 as common, and pin 8 as NO, so energizing any of these relays closes common-to-NO and applies the supply rail directly onto the node the low-side MOSFET is pulling toward ground. On K1/K2/K3/K5 this can short 12V_Main to GND_Main; on K4 it can short 12v_Safty to GND_Safty.

Recommendation

Separate the relay contact circuit from the coil-return node. Keep each MOSFET drain connected only to relay pin 12, and wire the external switched path only through the contact pins (pin 9 as common to the external circuit, and pin 8 or pin 10 as NO/NC as intended). Recheck the symbol/footprint pin mapping against the EE2 non-latch pinout before respin.

Affected Components

J8, J9, J14, J16, J17, K1, K2, K3, K4, K5, Q6, Q7, Q8, Q9, Q10

Affected Nets

12V_Main, 12v_Safty, GND_Main, GND_Safty, Net-(J8-Pad2), Net-(J9-Pad2), Net-(J14-Pad2), Net-(J16-Pad2), Net-(J17-Pad2)

Datasheet Evidence

EE2-12NUH.pdf (p.3)

"Coil terminals: Pins 1 and 12."

EE2-12NUH.pdf (p.3)

"Common (COM): Pins 4 and 9. Normally Closed (NC): Pins 3 and 10. Normally Open (NO): Pins 5 and 8."

2. [A2] [PWR-002] A2 pin 36 (3V3) is hard-tied to the external 3.3 V rail

Full Description

With A2 pin 36 connected directly to 3.3v_Safty, and A2 pins 39 (VSYS) and 40 (VBUS) left unconnected in this design, the Pico module is being powered by driving the node that the datasheet identifies as the on-board SMPS 3.3 V output. The Pico datasheet defines VBUS→VSYS as the module's input power path and defines 3V3 as the SMPS-generated supply. Because the Pico module also contains its own micro-USB-fed VBUS path, plugging in USB or otherwise powering the module through its documented input path would hard-tie the on-board 3.3 V source to the external 3.3v_Safty regulator with no OR-ing.

Recommendation

Power the Pico through VSYS as documented, or isolate pin 36 from the external 3.3 V regulator. If intentional 3.3 V injection is required, add a power-path scheme or otherwise guarantee that the module can never be powered from USB/VSYS at the same time.

Affected Components

A2

Affected Nets

3.3v_Safty

Datasheet Evidence

RaspberryPi_Pico.pdf (p.9)

"3V3 is the main 3.3V supply to RP2040 and its I/O, generated by the on-board SMPS."

RaspberryPi_Pico.pdf (p.20)

"VBUS is the 5V input from the micro-USB port, which is fed through a Schottky diode to generate VSYS."

RaspberryPi_Pico.pdf (p.20)

"VSYS is the main system 'input voltage' and feeds the RT6150 buck-boost SMPS, which generates a fixed 3.3V output"

3. [/MainControler/DataToSafty] [OD-001] U5 phototransistor output has no hardware pull-up on /MainControler/DataToSafty

Full Description

Given U5 pin 4 is connected to /MainControler/DataToSafty and that full-net inspection shows this net only reaches U5 pin 4 and U18 GPIO4, with no resistor or other pull-up present, the optocoupler output has no defined high state. TCMT1109 is a phototransistor-output device with pin 4 as the collector and pin 3 as the emitter, so when the LED is off the collector is simply open rather than actively driven high.

Recommendation

Add a pull-up resistor on /MainControler/DataToSafty to the main-side logic rail. If the intent is to rely on an MCU internal pull-up, document that choice and verify that the pull-up is enabled in all required reset/boot states; otherwise use an external resistor for a guaranteed default high level.

Affected Components

U5, U18

Affected Nets

/MainControler/DataToSafty

Datasheet Evidence

TCMT1109.pdf (p.1)

"Optocoupler, Phototransistor Output, Single Channel"

TCMT1109.pdf (p.1)

"Pin 4: Collector (C); Pin 3: Emitter (E)."

Warnings (8)

1. [D22] [PROT-001] SMAJ24CA clamp level exceeds the 35 V input-capacitor rating

Full Description

With D22 connected directly across 12V_Main and C12/C63 also tied across that same input rail, the SMAJ24CA has a 24 V reverse stand-off, breaks down at 26.7 V minimum / 28.1 V typical, and its specified maximum clamp reaches 38.9 V for a 10/1000 μ s surge or 50 V for an 8/20 μ s surge. That means a surge large enough to drive D22 near its datasheet clamp can overstress the 35 V bulk capacitors C12 and C63.

Recommendation

Use an input protection scheme whose worst-case clamp on 12V_Main stays below the capacitor voltage rating, or increase C12/C63 to a rating comfortably above the selected TVS clamp for the intended surge standard.

Affected Components

D22, C12, C63

Affected Nets

12V_Main

Datasheet Evidence

SMAJ24CA.pdf (p.3)

"SMAJ24A/CA ... VRM 24 V; VBR at IBR Min. 26.7 V Typ. 28.1 V"

SMAJ24CA.pdf (p.3)

"SMAJ24A/CA ... 10 / 1000 μ s VCL Max. 38.9 V ... 8 / 20 μ s VCL Max. 50 V"

2. [Q6/Q7/Q8/Q9] [FET-002] BSP296N is driven from 3.3 V logic without a 3.3 V RDS(on) guarantee

Full Description

Given /MainController/Relay1, /MainController/Relay2, /MainController/Relay3, and /SaftyProcessor/saftyRelay drive Q6/Q7/Q8/Q9 gates through 120 Ω resistors from 3.3 V logic, the BSP296N sees about 3.3 V gate drive. The datasheet gives guaranteed RDS(on) only at VGS = 4.5 V and 10 V; VGS(th) = 0.8 to 1.8 V is only the 100 μ A threshold test, not an on-resistance guarantee. The relay coil itself is light (12.0 V coil, 1,028 Ω nominal, about 11.7 mA), so this may still function, but the MOSFET is not formally characterized as a 3.3 V logic-level relay switch.

Recommendation

Use a MOSFET with RDS(on) explicitly specified at 2.5 V or 3.3 V gate drive, or validate worst-case relay pickup/release across tolerance and temperature with the BSP296N at a 3.3 V gate drive before relying on this path.

Affected Components

K1, K2, K3, K4, Q6, Q7, Q8, Q9, R36, R40, R64, R73

Affected Nets

/MainController/Relay1, /MainController/Relay2, /MainController/Relay3, /SaftyProcessor/saftyRelay, Net-(Q6-GATE), Net-(Q7-GATE), Net-(Q8-GATE), Net-(Q9-GATE)

Datasheet Evidence

BSP296NH6327XTSA1.pdf (p.2)

"Gate threshold voltage VGS(th), VDS=VGS V, ID=100 μ A: 0.8 min, 1.4 typ, 1.8 max."

BSP296NH6327XTSA1.pdf (p.2)

"Drain-source on-state resistance RDS(on): VGS=4.5 V, ID=0.95 A: 371 typ, 800 max m Ω ; VGS=10 V, ID=1.2 A: 329 typ, 600 max m Ω ."

EE2-12NUH.pdf (p.9)

"Nominal Coil Voltage 12.0 VDC; Coil Resistance 1,028.0 Ω $\pm$ 10%."

3. [U4/U5/U6] [CUSTOM] TCMT1109 LED drive current is below the datasheet CTR test condition

Full Description

With R12, R17, and R18 each 680 Ω from a 3.3 V GPIO into a TCMT1109 input LED, the LED current is only about $(3.3 \text{ V} - 1.6 \text{ V}) / 680 \Omega \approx 2.5 \text{ mA}$ using the datasheet VF max value, or about $(3.3 \text{ V} - 1.35 \text{ V}) / 680 \Omega \approx 2.9 \text{ mA}$ using the typical VF value. The TCMT1109 CTR grade for this part is specified as 200 % to 400 % at VCE = 5 V and IF = 5 mA, so these channels are being used below the datasheet's stated CTR test point and transfer margin is not guaranteed across tolerance, temperature, and aging.

Recommendation

Reduce R12/R17/R18 or choose an optocoupler with guaranteed CTR at the actual 2 mA to 3 mA drive level. After changing the LED drive, re-check the collector-side pull-up currents so each channel still has adequate logic-low margin.

Affected Components

U4, U5, U6, R12, R17, R18

Affected Nets

/SaftyProcessor/PicoTx, /MainController/DataFromSafty, /MainController/Fault, Net-(R12-Pad1), Net-(R17-Pad2), Net-(R18-Pad2)

Datasheet Evidence

TCMT1109.pdf (p.2)

"Forward voltage IF = 50 mA VF typ. 1.35 max. 1.6 V"

TCMT1109.pdf (p.3)

"VCE = 5 V, IF = 5 mA ... TCMT1109 ... MIN. 200 ... MAX. 400 %"

4. [/MainController/DataFromSafty] R34 pull-up forward-biases the optocoupler LED on /MainController/DataFromSafty

Full Description

With /MainController/DataFromSafty tied to R34 = 1 k Ω up to 3.3V_Main while also driving the TCMT1109 input LED through R17 = 680 Ω to GND_Main, the channel is not idle-off when GPIO5 is high-impedance. Using the TCMT1109 forward-voltage maximum of 1.6 V, the pull-up alone biases about $(3.3 \text{ V} - 1.6 \text{ V}) / (1 \text{ k}\Omega + 680 \Omega) \approx 1.0 \text{ mA}$ through the LED, so the isolated channel defaults partially on unless U18 actively pulls the net low.

Recommendation

Remove R34 from the LED-drive net, or move the pull-up to the phototransistor output side where the isolated receiver needs a defined high level. If this default-on, active-low behavior is intentional, document it and verify the receiving firmware is designed for that polarity from reset.

Affected Components

R17, R34, U4, U18

Affected Nets

/MainController/DataFromSafty

Datasheet Evidence

TCMT1109.pdf (p.1)

"Pin 1: Anode (A) of the infrared-emitting diode; Pin 2: Cathode (C) of the infrared-emitting diode"

TCMT1109.pdf (p.2)

"Forward voltage; IF = 50 mA; VF typ. 1.35 V, max. 1.6 V"

5. [Net-(FB2-Pad2)] MAX31856 negative thermocouple boundary network differs from the vendor reference schematic

Full Description

With J11 pin 1 routed through FB2 to a shared boundary node that then connects through R50 (2k) to U13 T- and through R92 (2k) to U13 BIAS, the negative thermocouple lead is split equally between the T- input and the BIAS pin. On the MAX31856 evaluation schematic, the corresponding BIAS-coupling resistor is labeled R1 = 20K while the T- link is labeled R2 = 0, so this design makes the BIAS path 10× stronger than the reference and removes the direct/0Ω T- connection.

Recommendation

Verify this boundary network against the MAX31856 IC pin-function requirements, especially BIAS/open-thermocouple behavior. If the intent is to follow the vendor reference circuit, connect the filtered thermocouple negative node directly or through 0Ω to T-, and couple BIAS to that same node with about 20kΩ instead of 2kΩ.

Affected Components

FB2, J11, R50, R92, U13

Affected Nets

Net-(FB2-Pad1), Net-(FB2-Pad2), Net-(U13-T-), Net-(U13-BIAS)

Datasheet Evidence

MAX31856.pdf (p.7)

"R1 20K"

MAX31856.pdf (p.7)

"R2 0"

6. [J20/J21/J22] Jack inputs are left undefined when no plug is inserted

Full Description

With J20/J21/J22 TIP connected only through R7/R8/R9 to the U1/U2/U3 non-inverting inputs, and with the SJ2-35813B-SMT tip-switch pins left unconnected, removing a plug leaves each monitored input without a defined DC reference. That can let the op-amp inputs and the ADC-held values drift or pick up noise instead of settling to a known 'no sensor/no signal' state.

Recommendation

Add a defined bias for each input, such as a pull-down/pull-up resistor on the jack TIP or op-amp input node. If you want the unplugged state to be deterministic without a permanent bias load, use pin 10 (tip switch) of J20/J21/J22 to force a known state or to provide plug-detect logic.

Affected Components

J20, J21, J22, R7, R8, R9, U1, U2, U3

Affected Nets

Net-(J20-TIP), Net-(J21-TIP), Net-(J22-TIP), Net-(D1-K), Net-(D2-K), Net-(D15-K)

Datasheet Evidence

SJ2-35813B-SMT-TR.pdf (p.2)

"2 tip"

SJ2-35813B-SMT-TR.pdf (p.2)

"10 tip switch"

7. [Net-(J22-TIP)] Same analog input is exposed on two external connectors in parallel

Full Description

With J22 pin 2 (tip) and J7 pin 2 hard-tied together on Net-(J22-TIP), the same analog field node is presented simultaneously on both the 3.5mm jack and the 2-position screw terminal. Any simultaneous attachment of two external devices would directly parallel them on this node, which can alter the measured signal or create source contention unless one connector is intended as a documented single-use or assembly-option input.

Recommendation

Document that only one connector is to be used/populated at a time, or make one connector a DNP/assembly option. If both connectors must remain fitted, add switching or isolation so two external devices cannot be hard-paralleled.

Affected Components

J22, J7

Affected Nets

Net - (J22-TIP)

Datasheet Evidence

SJ2-35813B-SMT-TR.pdf (p.2)

"PIN 1 sleeve 2 tip 3 ring 10 tip switch"

1935161.pdf (p.1)

"PCB terminal block ... Number of positions: 2, Connection method: Screw connection"

8. [J21] J21 uses a 3-conductor jack part number, but only tip/sleeve are implemented

Full Description

With J21 TIP_2 and J3 pin 2 hard-tied on Net-(J21-TIP) and J21 SLEEVE_1 tied to GND_Safty, the implemented interface only uses two contacts of connector J21. The selected SJ2-35813B-SMT-TR datasheet defines additional contacts for ring (pin 3) and tip switch (pin 10), but those contacts are not represented in the implemented interface. Any mating accessory expecting the ring conductor or jack switch to be defined will see those functions left uncontrolled.

Recommendation

If this interface is intended to be strictly 2-wire, use a true 2-conductor jack or clearly document and enforce the required mating plug type. If pin 3 and pin 10 are exposed in the symbol, explicitly connect them or mark them NC according to the intended interface.

Affected Components

J21

Affected Nets

Net-(J21-TIP), GND_Safty

Datasheet Evidence

SJ2-35813B-SMT-TR.pdf (p.2)

"2 tip"

SJ2-35813B-SMT-TR.pdf (p.2)

"3 ring"

SJ2-35813B-SMT-TR.pdf (p.2)

"10 tip switch"

Missing Information (7)

[IC1/IC3] LT8631 passive-value requirements could not be verified

The LT8631-specific requirements for the feedback dividers, RT resistors, soft-start capacitors, bootstrap capacitors, INTVCC capacitors, output capacitors, and inductor selections around IC1 and IC3 could not be verified because the LT8631 datasheet was not available.

Information Needed

LT8631 datasheet pages covering FB reference, frequency-programming resistor, soft-start, bootstrap/INTVCC capacitor requirements, recommended inductor/output-capacitor ranges, and stability constraints.

[/MainRegulator/PGoodMain] LT8631 PG/EN boundary loading around D3/R108 could not be confirmed

D3 cathode, IC1 PG, IC3 EN/UV, and divider R51/R52 all share /MainRegulator/PGoodMain, while D3 anode is fed from 12V_Main through R108 and R91. Whether that LED branch is compatible with the LT8631 PG output behavior and IC3 EN/UV limits could not be confirmed without the LT8631 datasheet.

Information Needed

LT8631 datasheet facts for PG pin output type and recommended pull-up behavior, plus EN/UV threshold, hysteresis, absolute maximum voltage, and bias-current limits.

[Net-(D5-K)] LT8631 PG-driven indicator load on D5/R110/R111 could not be verified

D5 is connected from IC3 PG to 12V_Main through R111 and R110, so this indicator depends on the LT8631 PG pin tolerating the pull-up voltage and sinking the resulting LED current. Those PG-pin limits could not be verified without the LT8631 datasheet.

Information Needed

LT8631 datasheet details for PG output type, absolute maximum pin voltage, recommended pull-up rail, and sink-current or VOL specifications.

[IC1/IC3] Grounded LT8631 NC and SYNC/MODE pin terminations remain unverified

IC1 and IC3 pins named NC_1, NC_2, NC_3, NC_4, and SYNC/MODE are tied to GND_Main in this design. The required pin handling could not be verified without the LT8631 datasheet.

Information Needed

LT8631 datasheet pin table and application guidance for NC pins and SYNC/MODE termination options.

[U13] MAX31856 input-filter and BIAS/T- requirements could not be fully verified

The available MAX31856 material showed the reference schematic values, but did not provide explicit electrical limits or recommended networks for the BIAS, T+, and T- pins. That leaves the acceptability of the implemented ferrite-bead, 2 k Ω series, and 2 k Ω BIAS-coupling network around U13 only partially verified.

Information Needed

MAX31856 datasheet or application guidance giving BIAS/T-/T+ pin behavior, open-thermocouple detection requirements, allowable series resistance or ferrites in the thermocouple leads, and any recommended external RC network.

[U13] Unable to confirm the required handling of MAX31856 pin 6 DNC

The available MAX31856 material did not include a pin-table or package-note statement for pin 6 DNC, so it could not be confirmed whether that pin must be left open or handled in a specific way.

Information Needed

MAX31856 full pin description or package note for pin 6 DNC.

[D1/D2/D15/D16/D17/D19] Differential clamp zener networks could not be fully verified

Verification was not possible for D1/D2/D15/D16/D17/D19 because no datasheet for BZX84C3V3LT1G was provided. That prevents confirming the exact clamp-voltage tolerance, leakage, dynamic resistance, and pin mapping of the back-to-back protection networks.

Information Needed

BZX84C3V3LT1G datasheet, especially breakdown-voltage tolerance, leakage current, dynamic resistance, and package pin mapping.

Verified Items (76)

Components (48)

J10	C10	R16	R89
R90	C25	C26	C27
C28	C29	J2	J3
J7	C50	C51	C52
D9	D10	D11	D18
D20	D23	R37	R41
R65	D25	D33	R74
R67	U7	U18	R20
R25	R26	R27	R28
R29	J1	R61	R62
R11	R31	R32	R33
D3	R108	J20	J22

Nets (28)

12V_Main	/SaftyProcessor/estop	/SaftyProcessor/thermoDrdy	/SaftyProcessor/thermoFault
3.3v_Safty	GND_Safty	Net-(Q9-GATE)	Net-(Q10-GATE)
/MainControler/SDA	/MainControler/SCL	/MainControler/CLK	/MainControler/MOSI
/MainControler/MISO	/MainControler/CS3	/MainControler/Fault	/MainControler/IO_Ex- pander_IRQ
/MainControler/IO_Ex- pander_RST	/SaftyProcessor/CLK	/SaftyProcessor/CS0	/SaftyProcessor/MISO
/SaftyProcessor/MOSI	/SaftyProcessor/Current1	/SaftyProcessor/Current2	/SaftyProcessor/Current3
Net-(D3-A)	Net-(D5-A)	Net-(J20-TIP)	Net-(J21-TIP)